Lecture 3 (*February 22*)

Reading: Sections 2.4–2.5
Viewing: Lecture 3

Let's delve a little deeper into matrix multiplication. The key thing is the following:

$$\Big(m \times n \textbf{ matrix}\Big)\Big(n \times p \textbf{ matrix}\Big) = \Big(m \times p \textbf{ matrix}\Big) \tag{15}$$

AB **only makes sense if the number of columns of** A **equals the number of rows of** B.

Assume A is an $m \times n$ matrix with entries a_{ij} and that B is an $n \times p$ matrix with entries b_{ij}. Then the product $C = AB$ is a $m \times p$ matrix, whose entry on the i–th row and j–th column is equal to:

$$\boxed{c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}} \tag{16}$$

Here's how to see the same computation by looking at the matrices themselves:

$$\underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & c_{ij} & * & * \\ * & * & * & * & * \end{bmatrix}}_{AB} = \underbrace{\begin{bmatrix} * & * & * & * \\ a_{i1} & a_{i2} & \dots & a_{in} \\ * & * & * & * \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} * & * & b_{1j} & * & * \\ * & * & b_{2j} & * & * \\ * & * & \dots & * & * \\ * & * & b_{nj} & * & * \end{bmatrix}}_{B}$$

(in the example above, $m = 3$, $n = 4$, $p = 5$, $i = 2$, $j = 3$). The i–th row of the matrix A and the j–th column of the matrix B are vectors of the same size, namely n. The only distinction is that the former vector is written horizontally and the latter vector is written vertically. Formula (16) precisely says that the (i, j)–entry in the matrix AB is the **dot product** of these two vectors.

What if we wanted to look at the entire i–th row of the matrix AB?

$$\underbrace{\begin{bmatrix} * & * & * & * & * \\ c_{i1} & c_{i2} & \dots & \dots & c_{ip} \\ * & * & * & * & * \end{bmatrix}}_{AB} = \underbrace{\begin{bmatrix} * & * & * & * \\ a_{i1} & a_{i2} & \dots & a_{in} \\ * & * & * & * \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} b_{11} & b_{12} & \dots & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & \dots & b_{2p} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & \dots & b_{np} \end{bmatrix}}_{B}$$

The entire i–th row of AB is a linear combination of the rows of B, according to the rule:

$$\begin{bmatrix} c_{i1} & c_{i2} & \dots & \dots & c_{ip} \end{bmatrix} = a_{i1}\begin{bmatrix} b_{11} & b_{12} & \dots & \dots & b_{1p} \end{bmatrix} + \cdots + a_{in}\begin{bmatrix} b_{n1} & b_{n2} & \dots & \dots & b_{np} \end{bmatrix}$$

Similarly, we may ask about the entire j–th column of the matrix AB:

$$\underbrace{\begin{bmatrix} * & * & c_{1j} & * & * \\ * & * & \dots & * & * \\ * & * & c_{mj} & * & * \end{bmatrix}}_{AB} = \underbrace{\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} * & * & b_{1j} & * & * \\ * & * & b_{2j} & * & * \\ * & * & \dots & * & * \\ * & * & b_{nj} & * & * \end{bmatrix}}_{B}$$

The entire j–th column of AB is a linear combination of the columns of A, according to the rule:

$$\begin{bmatrix} c_{1j} \\ \vdots \\ c_{mj} \end{bmatrix} = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} b_{1j} + \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} b_{2j} + \cdots + \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} b_{nj}$$

The two computations we have just seen are particular instances of **block multiplication** of matrices. Explicitly, we say that matrices A and B are written in **block form**:

$$A = \left[\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right] \quad \text{and} \quad B = \left[\begin{array}{c|c} B_{11} & B_{12} \\ \hline B_{21} & B_{22} \end{array}\right] \tag{17}$$

if A_{ij} is an $m_i \times n_j$ matrix, and B_{jk} is a $n_j \times p_k$ matrix for all $i, j, k \in \{1, 2\}$ and some natural numbers $m_1, m_2, n_1, n_2, p_1, p_2$. **Very important:** the sizes of the blocks in the block form of a matrix always match at the boundaries (for example, the number of rows of A_{11} is the same as the number of rows of A_{12}, and the number of columns of A_{11} is the same as the number of columns of A_{21}). Then the product matrix AB also has a block form:

$$AB = \left[\begin{array}{c|c} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ \hline A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right] \tag{18}$$

where each of the four blocks in (18) is matrix multiplication in its own right (the sizes of these blocks are $m_i \times p_k$). Thus, you may reduce the multiplication of the bigger matrices A and B to the multiplication of the smaller matrices A_{ij} and B_{jk}. This allows you to recursively reduce matrix multiplication to the smallest blocks, namely 1×1, which are simply individual matrix entries.

Example 2. *Block matrices such as* (17) *are the same thing as ordinary matrices, they are just represented (construed) differently. For example, consider the following matrices in block form:*

$$A = \left[\begin{array}{cc|c} 1 & -1 & 2 \\ \hline -4 & 1 & 0 \end{array}\right] \quad \textit{and} \quad B = \left[\begin{array}{ccc|c} 1 & -1 & 2 & -3 \\ -4 & 0 & 3 & -6 \\ \hline 1 & 4 & 0 & 2 \end{array}\right] \tag{19}$$

You could either think of A and B as 2×3 and 3×4 matrices (respectively) by deleting the various horizontal and vertical lines, or you could think of them as block matrices in the form (17)*, with:*

$$A_{11} = \begin{bmatrix} 1 & -1 \end{bmatrix} \qquad A_{12} = \begin{bmatrix} 2 \end{bmatrix}$$
$$A_{21} = \begin{bmatrix} -4 & 1 \end{bmatrix} \qquad A_{22} = \begin{bmatrix} 0 \end{bmatrix}$$
$$B_{11} = \begin{bmatrix} 1 & -1 & 2 \\ -4 & 0 & 3 \end{bmatrix} \qquad B_{12} = \begin{bmatrix} -3 \\ -6 \end{bmatrix}$$
$$B_{21} = \begin{bmatrix} 1 & 4 & 0 \end{bmatrix} \qquad B_{22} = \begin{bmatrix} 2 \end{bmatrix}$$

Then you may apply formula (18) *to compute the product of matrices AB by summing products of their constituent blocks A_{ij} and B_{jk}. Doing so, one obtains:*

$$AB = \left[\begin{array}{ccc|c} 7 & 7 & -1 & 7 \\ \hline -8 & 4 & -5 & 6 \end{array}\right]$$

One could also obtain the same result for AB by multiplying the matrices (19) *directly, just as you would multiply a 2×3 with a 3×4 matrix. Therefore, whether we think of matrices as being in block form or not, computations yield the same answer. So block form is just a matter of perspective.*

Matrix multiplication satisfies many important rules, such as:

- associativity: $A(BC) = (AB)C$
- distributivity: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
- unit element: $AI = IA = A$, where $I = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$ is called the **identity** (or unit) matrix. People sometimes write I_n instead of I if they wish to specify the identity matrix of size $n \times n$.

Given a <u>square</u> matrix A, we can form its powers:

$$A^p = \underbrace{AA \dots A}_{p \text{ times}}$$

Just like with numbers, we set:

$$A^0 = I = \text{the identity matrix}$$

but can we talk about negative powers? This is the point of the following definition.

Definition 1. *Given a square matrix A, its **inverse** is a matrix A^{-1} with the property that:*

$$\boxed{AA^{-1} = A^{-1}A = I} \tag{20}$$

Not all square matrices have inverses (those which do are called **non-singular** [3] and those which do not are called **singular**) but if an inverse exists, it is unique. For example, consider the matrix:

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 10 \end{bmatrix}$$

If it had an inverse $A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then we would have:

$$\begin{bmatrix} 1 & 5 \\ 2 & 10 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{21}$$

As we have seen, this implies that the columns of the identity matrix, namely:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{22}$$

[3] There is no ambiguity between this notion of "(non)-singularity" and the one from the previous Lecture, as we will soon see that the two notions are equivalent for square matrices.

are both linear combinations of the columns of A, namely:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 5 \\ 10 \end{bmatrix} \tag{23}$$

But this is clearly impossible, as the vectors (23) are both on the line of slope 2 in the plane, and therefore so is any linear combination of them. Meanwhile, the vectors (22) are not confined to this line, so we conclude that formula (21) cannot hold for any a, b, c, d, and therefore A is singular.

Gauss-Jordan elimination provides an algorithm for computing the inverse of a matrix A, if such an inverse exists. Start by constructing the augmented matrix:

$$\left[\begin{array}{c|c} A & I \end{array}\right] \tag{24}$$

where A and I are both $n \times n$. If A is non-singular, then Gauss-Jordan elimination will output:

$$\left[\begin{array}{c|c} I & B \end{array}\right] \tag{25}$$

for some matrix B. Now how does Gauss-Jordan elimination get you from (24) to (25)? We have seen last time that it does this by multiplying (24) on the left with another matrix C:

$$C \left[\begin{array}{c|c} A & I \end{array}\right] = \left[\begin{array}{c|c} I & B \end{array}\right]$$

But from the rules of multiplying block matrices, this is equivalent to the two matrix equalities:

$$CA = I \quad \text{and} \quad CI = B$$

Therefore, $C = CI = B$, hence $BA = I$, hence B is the sought-for inverse of A. So the right-most block of (25), which is produced by Gauss-Jordan elimination, is precisely the inverse of A. If Gauss-Jordan elimination does not produce the identity matrix in the left-most block of (25), which would be the case if there were rows without pivots at the end of the elimination algorithm, then A would not have an inverse (i.e. A would be singular).

Being able to calculate inverses of matrices is a very helpful skill, since they allow you to solve square systems of equations (i.e. systems where the number of equations is equal to the number of unknowns). Specifically, if you multiply a system $A\boldsymbol{v} = \boldsymbol{b}$ on the left by A^{-1}, then you obtain:

$$\boxed{\boldsymbol{v} = A^{-1}\boldsymbol{b}} \tag{26}$$

which is precisely the same as solving for the unknown vector $\boldsymbol{v}$. A useful tool in computing matrices is the following formula:

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}} \tag{27}$$

which says that the inverse of a product (of square matrices) is the product of the inverses, but taken in reverse order. This is very meaningful, since the order in which we multiply matrices is important (a fancy way of saying this is that matrix multiplication is not a commutative operation). Finally, a nice consequence of being able to define inverse matrices is that the expression:

$$A^n \tag{28}$$

now makes sense for any integer n, if A is a non-singular square matrix. If n is negative, then you just define (28) as the $-n$ th power of the inverse matrix A^{-1}. This is just the basic requirement that the usual formulas for powers:

$$A^m A^n = A^{m+n} \qquad \text{and} \qquad (A^m)^n = A^{mn} \tag{29}$$

continue to hold for matrices, as they do for numbers.